

Reading Support: Degrees of String Admissibility

Companion to Week 5

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The hierarchy classifies formal languages

Supports: The classical hierarchy is a chain of set inclusion

A formal language is a set of strings over an alphabet. The Chomsky hierarchy groups such sets according to restrictions on the grammars that describe them. Regular languages form a proper subset of the context-free languages, which sit within successively more inclusive families.

This classification concerns descriptive resources. It does not rank natural languages or their speakers by complexity.

MOLR §3.1

Finer families locate linguistic dependencies more accurately

Supports: Natural-language work cuts the hierarchy at finer seams

The classical hierarchy leaves large intervals between its families. Linear languages refine the region between regular and context-free languages. Linear-indexed languages refine the region between context-free and context-sensitive languages.

These families allow a description to preserve more than a weaker formalism without admitting everything available to a much stronger one.

MOLR §3.1

A pumping lemma supplies a necessary condition

Supports: A pumping lemma states a necessary property of a family

A pumping lemma says that every sufficiently long string in a language belonging to the family has a decomposition whose selected parts can be repeated while remaining in the language.

To prove nonmembership, we show that every decomposition allowed by the lemma fails for some repetition. Exhibiting a convenient decomposition or one pumpable string cannot prove membership.

MOLR §3.2

Regular pumping repeats one bounded region

Supports: Regular admissibility permits one pumped region

If L is regular, then for some pumping length n , every $z \in L$ with $|z| \geq n$ can be written uvw , where $|v| \geq 1$, $|uv| \leq n$, and $uv^i w \in L$ for every $i \geq 0$.

The repeated material lies within a bounded initial portion. A language requiring an arbitrarily distant matched change can defeat this condition.

MOLR §3.2

Context-free pumping coordinates two regions

Supports: Context-free admissibility permits two coordinated regions

If L is context-free, every sufficiently long z admits a decomposition $uvwxy$, with $|vx| \geq 1$, $|vwx| \leq n$, and $uv^iwx^iy \in L$ for every $i \geq 0$.

The two pumped substrings can express coordinated growth. Their bounded combined neighborhood reflects the local derivational source of that coordination.

MOLR §3.2

Linear pumping places a stronger restriction on the pair

Supports: Linear admissibility restricts where the paired growth occurs

Linear languages also permit a decomposition $uvwxy$ with v and x pumped together. Their condition restricts the material outside the central continuing dependency, corresponding to derivations with at most one nonterminal on each right-hand side.

The family can coordinate two regions while excluding independent context-free branches.

MOLR §3.2

Admissibility properties do not reify a processing device

Supports: Admissibility is structural without reifying a machine

Pumping arguments characterize strings through invariant decomposition properties. They do not claim that a speaker repeats substrings, searches decompositions, or stores a pumping length.

The classification tells us what degree of structural dependency a description must preserve. Grammars and automata later provide distinct ways to describe and compute strings with that property.

MOLR §3.3